

Human Development Index (HDI) Predictor

Code Layout, Readability & Reusability

Project: Human Development Index (HDI) Predictor

Phase: Project Development Phase

1. Folder Organization

The project is split into three clearly-scoped folders: Training/ (data generation and model fitting, run once), Flask/ (the deployable web app, model artifacts, and templates), and Dataset/ (data and EDA outputs).

2. Path Resolution for Reusability

app.py resolves file paths relative to its own location rather than the current working directory, so it runs correctly whether launched as "python app.py" from inside Flask/, or as "python Flask/app.py" from the project root:

```
BASE_DIR = os.path.dirname(os.path.abspath(__file__))
MODEL_PATH = os.path.join(BASE_DIR, 'HDI.pkl')
ENCODER_PATH = os.path.join(BASE_DIR, 'country_encoder.pkl')
```

3. Readability Practices

- Each script is commented section-by-section, matching the project's epic structure
- Descriptive variable names (x_train, y_pred, country_encoder) instead of single letters where clarity matters
- Try/except error handling in the /predict route with a clear fallback message on bad input

4. Reusability

- generate_dataset.py and train_model.py can be re-run independently to regenerate data or retrain the model
- The saved LabelEncoder ensures any new country added to the dataset is encoded consistently at both training and inference time
- HTML templates use Jinja2 templating (e.g. {% for c in countries %}) so the country list updates automatically if the dataset changes